

MnPASS Modeling and Pricing Algorithm Enhancement Task 2 Report: Lane Choice Model Development

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Executive Summary

This report describes the formulation and estimation of models which can be used to predict MnPASS transponder ownership by households in the Minneapolis–Saint Paul metropolitan area, and to predict the choices made by individual MnPASS subscribers regarding whether to use the MnPASS lane or the general purpose (GP) lanes while travelling on a MnPASS corridor. The purpose of this task is to develop a framework of models that can be incorporated into a microsimulation of travel behavior that will take place in a subsequent phase of the MnPASS Modeling and Pricing Algorithm Enhancement project.

The decision to subscribe to MnPASS and the decision to use a MnPASS lane can both be regarded as binary choices, with a particular relationship between them: only travelers who have previously made the choice to subscribe to MnPASS are presented with the later choices to actually use a MnPASS lane. For this reason both models are necessary in order to provide sufficient detail for microscopic modeling of travel behavior on MnPASS corridors.

The modeling process is informed by previous literature from the field of transportation choice analysis and regarding travelers' behavior responses to high-occupancy/toll (HOT) lanes. Construction of the models requires data regarding MnPASS subscribers and their behavior, which is provided by MnDOT. Additionally, this analysis makes use of broader demographic data as well as information describing travel times throughout the regional road and highways system.

Both models are specified using a binomial logit form, where the likelihood of choosing one alternative is a function of the benefits offered by the chosen alternative and the benefits offered by the rejected alternative. Though similar in form the models differ in scope; the transponder ownership model is based on location and aggregate demographic parameters while the lane choice model is specified using travelers' individually-experienced previous travel times.

The results provide a reasonable fit to the data and provide a framework that can be used in subsequent tasks and research projects that seek to understand the behavior and economics of HOT lane use.

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1 Introduction

This report describes the formulation and estimation of models which can be used to predict MnPASS transponder ownership by households in the Minneapolis–Saint Paul metropolitan area, and to predict the choices made by individual MnPASS subscribers regarding whether to use the MnPASS lane or the general purpose (GP) lanes while travelling on a MnPASS corridor. The purpose of this task is to develop a framework of models that can be incorporated into a microsimulation of travel behavior that will take place in a subsequent phase of the MnPASS Modeling and Pricing Algorithm Enhancement project.

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The modeling process is informed by previous literature from the field of transportation choice analysis and regarding travelers' behavior responses to high-occupancy/toll (HOT) lanes. This literature is briefly reviewed in [Section 2](#).

Construction of these models requires data regarding MnPASS subscribers and their behavior, which is provided by MnDOT. Additionally, this analysis makes use of broader demographic data as well as information describing travel times throughout the regional road and highways system. [Section 3](#) provides a discussion of the data sources used.

[Section 4](#) provides a detailed discussion of the methodology used to construct the models.

2 Background

2.1 Modeling Transportation Choices

In transportation analysis, travel time has historically been considered the primary user cost that drives decisions. However, it is important to recognize that, at the time of making a travel decision, travelers typically have (far) less than perfect information about what travel times will be. [18] Bovy and Stern [3] discuss the important role that previous experience plays in guiding travelers' choices, by setting expectations that, in turn, can guide decisions. In the field of route choice, this typically takes the form of recollections of previous travel times as well as the variation in those travel times.

Expected travel time is one type of time cost that travelers pay. But they also face another type of cost: uncertainty. Variation of travel time plays a significant role in current travel behavior research. "Variation" typically brings to mind metrics like variance and standard deviation, both of which indicate variation relative to the mean. However, De Palma and Picard [7] note several situations in travel analysis where the use of models based on variance or standard deviation is inconsistent with the expected traveler behavior. For example, arriving 10 minutes early and 10 minutes late are indistinguishable in terms of standard deviation, even though travelers might have a strong preference for one over the other [17]. Brownstone and Small [4] propose that avoiding unexpected lateness is far more important to travelers than avoiding earliness, and so construct models where the travel time variation is represented by the difference between the 90th and the 50th percentile travel times — implying that travelers focus on avoiding the worst case when making decision, rather than finding the best case.

2.2 High-Occupancy/Toll Lanes

HOT lanes with transponder-based toll collection are a relatively young transportation technology, and data collection methods have become established only over the past decade. Li et al. [9] and Li et al. [10] provide a template for analyzing HOT lane usage and behavior using two important sources of data: toll collection records, which capture every interaction that subscribers have with the HOT lane system, and GPS-based surveys which capture detailed trip logs, regardless of whether a HOT lane is used. To fully understand traveler behavior in a HOT lane system both must be used, since toll collection records alone cannot reveal traveler behavior when a HOT lane is *not* used.

Burris and Stockton [5] highlight the importance of understanding the economics and behavior associated with HOT lanes. They chronicle the results of several variations of HOV and HOT policies in Houston, where HOT lane pricing was implemented as a tool to more effectively guide an appropriate level of demand into the lane, while preserving travel time reliability. This importance is compounded by the fact that traveler behavior in HOT lane systems is often counterintuitive, as illustrated by Liu et al. [11], where higher HOT lane prices were found to increase demand.

Goodall and Smith [8] apply transportation choice theory in the context of HOT lanes. Their findings suggest that travel time reliability, rather than expected travel time savings, drives both the choice to enroll in HOT lane programs and the choice to use a HOT lane on a per-trip basis.

3 Data

3.1 MnPASS Subscriber Locations

Information describing the home location of MnPASS subscribers is collected in the form of billing addresses associated with each MnPASS account. This information is initially provided by MnPASS users when an account is created, and is of necessity updated when credit card billing addresses change. It is important to note that multiple vehicles may be associated with a single account, and that there is no requirement that the billing address be a home address — or that the drivers of the associated vehicles be related to the holder of the account. For example, a single MnPASS subscriber might use her business address as the billing address, or a business might create a MnPASS account using a single billing address to request multiple MnPASS transponders for a fleet of vehicles [14]. These facts have important implications for the use and interpretation of MnPASS subscriber location information.

To support this research project, MnDOT provided two datasets derived from the billing addresses of MnPASS subscribers. One dataset associates each MnPASS account number with the zip code of the account’s billing address; the second provides a count of MnPASS account billing addresses that fall within each Transportation Analysis Zone (TAZ) as defined by the Metropolitan Council for 2010. Additionally, the transponder logs collected during MnPASS operation provide both the transponder ID and the account ID for each logged MnPASS use. This allows the inference of an association between multiple transponder IDs and a single MnPASS account, though it is not guaranteed to provide a complete picture: some transponders might not be used during the data collection period.

Due to the size of zip code areas, the zip code association for each account does not provide a detailed enough geographic resolution for modeling that involves demographic characteristics and travel behavior. TAZs, especially as defined for 2010, describe far smaller geographic areas that are much better suited for analysis using demographic and travel behavior characteristics. Therefore, the count of MnPASS account addresses per TAZ is the primary locational dataset used in this analysis.

3.2 Regional Demographics

Demographic information describing the residents of the Minneapolis–Saint Paul metropolitan area is used to associate aggregate demographic characteristics with geographic areas for use in modeling transponder ownership, as well as to provide a statistical basis for constructing a synthetic population that will be used in subsequent research tasks to inform agent-based microsimulations.

This analysis uses demographic data collected by the U.S. Census Bureau’s American Community Survey (ACS). Specifically, block group-level estimates for 2007–2011 are used in modeling MnPASS transponder ownership, and block group-level estimates for 2006–2010 are used in the construction of a synthetic population. As of this writing, limitations in the PopGen software package prevent its use with Census data more recent than 2010 [1, 2].

3.3 Traveler Behavior

While the general structure of the lane choice model, which predicts whether an individual traveler will use a HOT lane or a general-purpose lane, is guided by existing literature and practice, the specific variables and parameters included in the model are estimated based on actual traveler behavior. The MnPASS system itself does not collect sufficient information to estimate this model: by design, it only records instances where subscribers choose to use a HOT lane. If a particular transponder ID is not recorded on a given day, there is no way to determine whether that traveler chose *not* to use a HOT lane, or simply was not exposed to the choice — for example, by not travelling or by using only routes which did not include a HOT lane option.

More robust model estimation is possible when the data includes observations of the same traveler making both choices [16]. Therefore, the lane choice model described in this report is estimated using data collected during a GPS-enabled travel behavior study conducted at the University of Minnesota in 2010. This study tracked the actual commute behavior of volunteer research subjects who were identified as likely to use the I-394 corridor in their daily commute. GPS traces and MnPASS transponder records were combined to provide a comprehensive view of each traveler's behavior — and in particular, the choice of whether to use the HOT lane on each individual trip. [6]

3.4 Accessibility

Transportation choice theory suggests that the decision to subscribe to MnPASS is based in part on the benefits that the MnPASS system provides to its subscribers. Here, this benefit is realized as the increase in accessibility to jobs that residents experience when they gain access to MnPASS lanes. To quantify the benefit, this analysis uses accessibility data calculated as part of MnDOT's Access to Destinations research project, described in Owen and Levinson [15]. Additionally, the method used in that project is employed to calculate accessibility after setting the speeds of MnPASS lanes to free-flow. The difference between these accessibility values is interpreted as the accessibility benefit that a potential subscriber would derive from being able to access the MnPASS system from a home location.

4 Methodology

4.1 Transponder Ownership Model

The transponder ownership model predicts whether a given household has access to a MnPASS account. The model is constructed as a binomial choice: a household can either have a MnPASS account, or no MnPASS account. This choice is modeled using the binomial logit form, where the likelihood of an individual making each choice depends on the utility provided by each. In turn, the utility of each choice is modeled as a function of a specific set of parameters.

4.1.1 Parameters

The transponder ownership model includes parameters which fall into two distinct classes: those which describe properties of a household's location (specifically, the TAZ in which it falls), and those which describe demographic properties of the households themselves. These parameters are summarized in [Table 1](#).

Table 1: Parameters used in transponder ownership model

Variable	Description	Mean	S.D.
D_e	Distance to MnPASS entry point (km)	16.18	11.93
D_m	Distance to downtown Minneapolis (km)	22.47	22.23
A_d	Accessibility benefit (1,000 jobs)	13.86	32.58
A_r	Accessibility benefit (ratio)	0.98	0.05
H_s	Average household size	2.58	0.50
H_i	Average household income	\$93,686	\$41,652
H_v	Average number of vehicles per household	1.91	0.47

Locational parameters

- **Distance to MnPASS entry point** (D_e) reflects the distance (in kilometers) from a household's TAZ to the nearest MnPASS entry point. Here, a MnPASS entry point is defined as a freeway entrance ramp that could be used to access a MnPASS lane for a trip towards downtown Minneapolis. Distances are measured along the road network.
- **Distance to downtown Minneapolis** (D_m) reflects the distance (in kilometers) from a household's TAZ to downtown Minneapolis. Distances are measured along the road network.
- **Accessibility benefit** (A_d , A_r) reflects the increase in accessibility to jobs that the MnPASS lane provides to the households TAZ. Accessibility to jobs is defined as the number of jobs that are reachable from a TAZ within 30 minutes during the AM peak period. This metric

is calculated using average GP lane speeds as measured by loop detectors, and then again using a free-flow speed for freeway segments where a MnPASS choice is available. The relationship between these numbers indicated the accessibility benefit that local residents could realize by gaining access to the MnPASS lanes. This analysis tests both the difference in accessibility (A_d) and the accessibility ratio between the two values (A_r).

Demographic parameters

- **Average household size** (H_s) reflects the size (number of persons) of the households in a TAZ.
- **Average household income** (H_i) reflects the total household income of the households in a TAZ.
- **Average number of vehicles per household** (H_v) reflects the number of vehicles available per household in a TAZ.

Additionally, the model includes an alternative-specific constant (ASC) to capture the variation in transponder ownership that is not explained by the above parameters.

4.1.2 Data Inclusion Based on Account/Household Ratio

As discussed in [Section 3.1](#), there are two important considerations to address when working with MnPASS account billing locations. First, there may be multiple transponders associated with a single account. Second, there is no guarantee that the account billing address reflects the home location of the account holder, or that the account holder is an individual rather than a business. These are addressed in two ways.

First, the transponder ownership model is implemented at the household level rather than the individual level. The model predicts whether a given household has access to a MnPASS account. In subsequent tasks of this research project, a simulated traveler who is associated with a MnPASS-capable household will be considered able to use HOT lanes while traveling.

Second, the model estimation excludes TAZs where the number of MnPASS accounts is greater than 50% of the number of households. Out of 2,915 TAZs with at least one household, this occurs only in 20 (0.6%), where the ratio of MnPASS accounts to household is 1.46 standard deviations or more above the mean. Among these outliers, it is reasonable to conclude that a significant number of MnPASS account billing addresses are not home addresses. Block groups with no households are also excluded. This is not an ideal method: it excludes some household accounts in TAZs with many business accounts, and it includes some business accounts in TAZs where the account/household ratio is low. However, it makes the best use of the available account location data.

4.2 Lane Choice Model

The lane choice model provides an approximation of the decision-making process that individual travelers go through when presented with the choice to use a HOT lane versus a GP lane while

traveling on a MnPASS corridor. As in the transponder ownership model, a binomial choice construction is used, and the choice is modeled using the binomial logit form.

4.2.1 Parameters

The lane choice model includes three parameters that are expected, based on theory and existing literature, to have a significant relationship to the likelihood of choosing to use a HOT lane. These parameters are summarized in [Table 2](#).

Table 2: Parameters used in lane choice model

Variable	Description
T_e	Expected travel time
V_s, V_r, V_q	Travel time variability
C	Expected toll cost

- **Expected travel time** (T_e) reflects the traveler's perception of the most likely travel time, in minutes, that would be experienced by choosing each option. The model implementation allows for the value to be determined either by the traveler's previous experience, or by a current posted travel time at the time of the choice, such as those provided by some highway message signs.
- **Travel time variability** (V_s, V_r, V_q) reflects the degree to which a traveler would expect the actual travel time to vary if the same choice was made repeatedly in the same conditions. Higher values indicate less reliable travel times. The model implementation allows for the use of any measure of travel time variability; this analysis tests three different measures based on Tilahun and Levinson [17] as well as Carrión [6]:
 1. (V_s) The standard deviation of previously-experienced travel times,
 2. (V_r) The right range of the distribution of previously-experienced travel times (the difference between the 90th and the 50th percentile values), and
 3. (V_q) The interquartile range of the distribution of previously-experience travel times (the difference between the 75th and the 25th percentile values).
- **Expected toll cost** (C) reflects the amount, in dollars, that the traveler expects to pay for each choice. In the case of the HOT lane choice, this represents the amount charged for using the MnPASS lane. The model implementation allows for the value to be determined either by the traveler's previous experience or by the current posted price at the time of the choice. In the case of the GP lane choice, this value is always zero.

Additionally, the model includes an alternative-specific constant (ASC) to capture the variation in HOT lane use likelihood that is not explained by the above parameters.

4.3 Model Evaluation

In order to evaluate the several models that will be presented, it will be necessary to evaluate how well each model fits the data. Logit models require specialized evaluation techniques. Model fit will be evaluated using the pseudo- R^2 popularized by McFadden [12] and used widely in transportation-related choice modes. In an evaluation of various pseudo- R^2 and related measures, Menard [13] identified several desirable qualities of McFadden's R^2 . Most importantly, it was found to be an “intuitively reasonable interpretation as a proportional reduction in error measure.” In the remainder of this report, McFadden's pseudo- R^2 will be denoted as “ pR^2 .” pR^2 values range from 0 to 1.0; a value of 0 indicates that a model provides no better description of the data than the null model, while a value of 1.0 indicates that a model describes the data perfectly.

It is important to note that the pR^2 measure is not directly comparable to R^2 as applied to linear regression models, because the pR^2 measure compares a specified model against a hypothetical null model, rather than directly against the data.

5 Results

5.1 Transponder Ownership Model

Two models were tested, using different implementations of the accessibility benefit parameter. Model T1 represents this benefit using the absolute increase in accessibility provided by the MnPASS lane, while model T2 uses the ratio of the MnPASS and the no-MnPASS accessibilities. Table 3 summarizes the results of the transponder ownership models. Of the two models, model T2 provides a better fit for the data. All of the parameters, both locational and demographic, are highly significant.

Table 3: Transponder ownership model

	Model T1	Model T2
Parameter	Coefficient	Coefficient
D_e	-0.142***	-0.142***
D_m	0.078***	0.072***
A_d	0.004***	—
A_r	—	-0.038***
H_s	-0.472***	-0.508***
H_i	0.011***	-0.011***
H_v	0.492***	0.522***
ASC	-4.95***	-0.999***
pR^2	0.628	0.635
n	2,350	2,350

*** Significant at 0.1% level or better

** Significant at 1.0% level or better

The two distance parameters D_e (distance to nearest MnPASS entrance) and D_m (distance to downtown Minneapolis) provide insight into the geographic aspects of MnPASS subscribership. It is intuitive that D_e has a negative coefficient, indicating that the likelihood of a household having a MnPASS subscription goes down as distance from a MnPASS entrance increases. Households which live near MnPASS entrances are more likely to find that MnPASS lanes provide a plausible route for more trips, and because less distance must be traveled before entering an MnPASS lane, MnPASS lanes may make up a larger proportion of an average trip.

D_m provides an interesting contrast — its coefficient is negative, indicating that as distance from downtown Minneapolis increases, households are more likely to have a MnPASS subscription. This makes sense when one considers the relative importance of MnPASS lanes to two commuters living different distances away from downtown Minneapolis. The commuter living farther away would spend longer driving on a MnPASS corridor, and so subscribing to MnPASS could improve

both travel time and reliability for a large percentage of his trip, and of his day. For the commuter living closer to downtown Minneapolis, time spent on a MnPASS corridor would be shorter, and so a subscription would offer less benefit.

The accessibility ratio coefficient indicates that as the ratio of non-MnPASS accessibility to MnPASS accessibility goes up, a household is more likely to subscribe to MnPASS. Since non-MnPASS accessibility cannot be higher than MnPASS accessibility, the maximum ratio value is 1, and lower values indicate a greater difference between the two accessibilities — indicating that the MnPASS lane offers a large accessibility benefit. Accordingly, households are more likely to subscribe to realize larger potential benefits.

The demographic parameters suggest that household size, vehicle availability, and income all play a role in determining subscription likelihood. H_i and H_v are relatively straightforward. As household income increases, subscription likelihood increases — which is consistent with an increasing value of time and of reliability. As the average number of vehicles per household increases, the likelihood that a household subscribed to MnPASS also increases. A greater number of cars suggests a greater importance of car travel.

H_s is less predictable: as average household size increases, subscription likelihood decreases. This is somewhat counterintuitive; one might expect that larger households would value the increased travel time reliability offered by MnPASS. It is possible that household size is acting as a proxy for another demographic or geographic variable. For example, workers from large households might seek jobs closer to home, and therefore would benefit less from the high-speed commutes offered by MnPASS.

5.2 Lane Choice Model

Three separate models are tested, each using a different metric for representing the expected variation of travel time. In all models, the variation is based on the set of previously-experienced travel times. Model L1 uses the standard deviation; model L2 uses a shortened right range of the distribution (90th – 50th percentile); and model L3 uses the interquartile range of the distribution (75th – 25th percentile). [Table 4](#) summarizes the results of the lane choice models. Of the three models, model L2 provides the best fit for the data. In this model, all of the parameters are significant at the 1% level or better.

In the lane choice models, all of the parameters represent costs — so it is no surprise that all coefficients are negative. As the cost of a choice increases, the likelihood of choosing it decreases.

As discussed in [Section 3](#), coefficients for this lane choice model were estimated using data collected during an earlier GPS survey of commute behavior. The values reported here will be used as initial seed values for calibration using the microsimulation developed in a later task of this research project. Therefore these values should be regarded as preliminary, though the model form is final.

Table 4: Lane choice model

Parameter	Model L1 Coefficient	Model L2 Coefficient	Model L3 Coefficient
T_e	-0.243***	-0.672***	-0.446***
V_s	-0.390***	—	—
V_r	—	-0.288***	—
V_q	—	—	-0.280***
C	-3.91***	-6.94***	-5.29***
ASC	0.0421	-2.23**	-1.67
pR^2	0.654	0.864	0.810
n	2,350	2,350	2,350

*** Significant at 0.1% level or better

** Significant at 1.0% level or better

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